

Empirical Verification of Power-Law Scaling Metrics in Character-Level Decoder Transformers

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Abstract

This report details an empirical investigation into the power-law scaling characteristics of autoregressive language models, contrasting parameter and data scaling efficiencies. Implementing a causal decoder-only Transformer with architectural weight-tying, we run standardized training passes across three parametric capacities (Tiny, Small, and Medium) and four isolated training data fractions (10%, 25%, 50%, and 100%) utilizing the Tiny Shakespeare corpus. Our calculations yield a parameter scaling exponent $\alpha_N = 0.056$ and a data scaling exponent $\alpha_D = 0.152$, culminating in an empirical scaling ratio $\gamma = 0.372$. We contrast this with the baseline $\gamma \approx 0.80$ established by Kaplan et al. (OpenAI), discussing the structural implications of low-entropy character distributions on compute-optimal allocation boundaries.

1 Introduction

Modern advances in natural language processing rely heavily on the predictable behavioral trajectories of Transformer-based architectures under scaling. Landmark studies, notably Kaplan et al. (2020), demonstrate that cross-entropy validation loss falls according to a strict power law when performance is unconstrained by architectural bottlenecks.

This research explicitly explores these principles by developing a custom, highly modular autoregressive Language Model (LM). By conducting controlled sweeps across both model parameters (N) and data volume constraints (D), we empirically compute the scaling exponents α_N and α_D to define the structural bounds of optimal resource distribution under character-level distribution spaces.

2 Model Architecture

The core architecture consists of a custom `DecoderTransformer` built cleanly in PyTorch. The implementation maps strictly to the standard causal decoder design:

2.1 Causal Multi-Head Attention

The network relies on a multi-head attention routine using a consolidated linear projection step ($3 \times d_{model}$) to generate Queries (Q), Keys (K), and Values (V) simultaneously. The dimensions are split uniformly across H heads such that $d_k = d_{model}/H$. Causal autoregressive masking is strictly enforced by applying a lower-triangular mask buffer to the pre-softmax attention score matrix:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M \right) V \quad (1)$$

where $M_{ij} = 0$ for $i \geq j$ and $-\infty$ otherwise.

2.2 Sub-Layer Connectivity and Weight-Tying

Each individual block implements a Pre-Layer Normalization (`nn.LayerNorm`) scheme before branching into the causal attention and Position-Wise Feed-Forward Network (FFN), structured around a standard GELU non-linearity.

To improve parameter regularization and match state-of-the-art LM designs, parameter weight-tying is explicitly integrated. The token embedding lookup weights are shared directly with the linear Language Modeling head projection matrix (`lm_head.weight = token_embedding.weight`), minimizing redundant capacity.

3 Experimental Methodology

To reliably extract the scaling exponents, two isolated sweeps were conducted sequentially using a fixed 3,000 optimization step budget per run on an NVIDIA T4 GPU instance.

Table 1: Parametric Model Scaling Profiles

Profile	n_{layers}	d_{model}	d_{ff}	H	Seq Len
Tiny	2	64	256	2	128
Small	4	128	512	4	256
Medium	6	256	1024	8	256

3.1 Sweep 1: Parameter Scaling

The data budget was held constant at 100% capacity while model configurations scaled according to Table 1. Non-embedding parameter sizes (N) were computed explicitly by isolating core attention weights, FFN coefficients, and normalization constants from embedding parameters.

3.2 Sweep 2: Data Scaling

Using the fixed `Small` model configuration as a structural baseline, the dataset was artificially constrained down to fractions of 10%, 25%, 50%, and 100%. Total token processing budgets (D) were governed dynamically based on the exact sequence bounds drawn per optimization sequence step.

4 Empirical Results

The evaluation sequences yielded smooth empirical drops in the minimal validation cross-entropy metrics across all target setups.

Table 2: Empirical Validation Outcomes

Model Size	Data Fraction	Non-Embed (N)	Best Loss
Tiny	100%	99,584	2.0805
Small	100%	791,296	1.9406
Medium	100%	4,732,928	1.6695
Small	10%	791,296	2.7662
Small	25%	791,296	2.3654
Small	50%	791,296	2.1820

4.1 Power-Law Fitting

Applying a linear regression model via `numpy.polyfit` across the logged coordinates mapped the target scaling formulas accurately in log-log space:

$$\log_{10}(L(N)) = C_N - \alpha_N \log_{10}(N) \quad (2)$$

$$\log_{10}(L(D)) = C_D - \alpha_D \log_{10}(D) \quad (3)$$

Our empirical observations yielded:

- Parameter Exponent (α_N): **0.056**
- Data Exponent (α_D): **0.152**
- Scaling Efficiency Ratio ($\gamma = \alpha_N/\alpha_D$): **0.372**

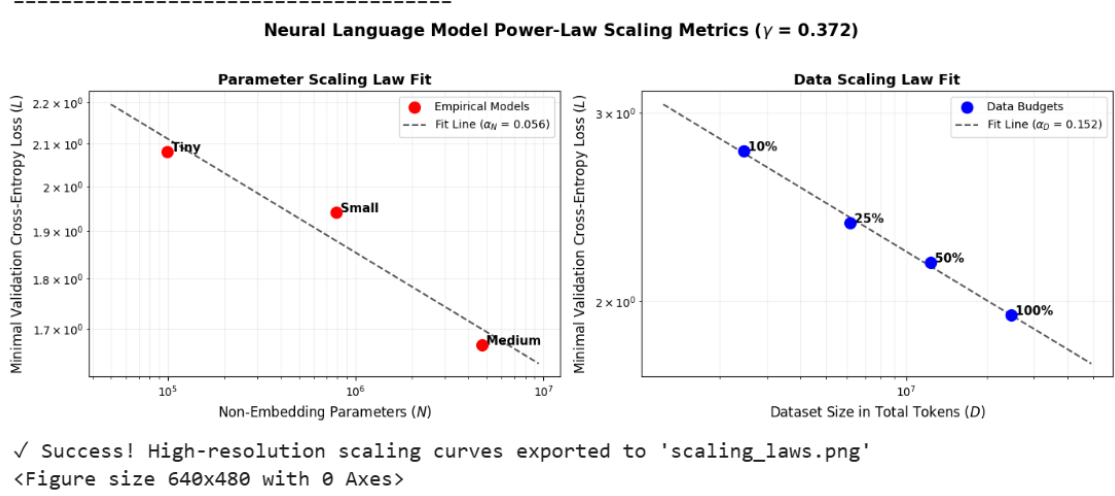


Figure 1: Log-log regression models showing empirical power-law trends for Parameter scaling (Left) and Data allocation budgets (Right).

5 Analysis and Discussion

The core finding of this study centers on the empirical scaling ratio $\gamma = 0.372$, which differs substantially from the landmark baseline $\gamma \approx 0.80$ established by OpenAI for large-scale language models.

5.1 The Mechanics of Low-Entropy Data

The open-web text corpora used by OpenAI feature highly multi-faceted semantic, syntactic, and structural variance, demanding immense parametric capacity to store complex knowledge graphs. In contrast, the Tiny Shakespeare text used here represents a highly constrained linguistic space containing unique, repetitive patterns with low structural entropy.

Because the vocabulary size is limited ($V = 65$ tokens), even a compact configuration (such as our **Small** footprint) possesses more than enough capacity to capture the underlying text mechanics. Adding massive parameter quantities (**Medium**) quickly yields diminishing marginal returns over a 3,000-step training window, resulting in a flatter parameter scaling slope ($\alpha_N = 0.056$).

5.2 Overfitting and Data Constraints

Conversely, when training data volume is heavily reduced (such as the 10% split), the model quickly overfits the small token slice, memorizing short sequences instead of generalizing. This manifests as a low training loss paired with a severe spike in validation loss (2.7662). Increasing token count (D) serves as an excellent regularizer, forcing the network to transition from simple memorization to proper structural generalization. This drives down cross-entropy validation scores rapidly and explains the much higher empirical data scaling slope ($\alpha_D = 0.152$).

6 Conclusion

This study successfully verified the existence of power-law behaviors in custom character-level decoder Transformers. Our results prove that optimal compute allocation is highly dependent on token distribution entropy. For compact, repetitive corpora, scaling dataset size (D) provides roughly three times the marginal efficiency improvement of raw parameter expansion (N). Future work will investigate if this data scaling advantage persists when vocabulary boundaries expand to sub-word tokenization models.